

Successive Differentiation, Leibnitz's Theorem, Expansion of Functions by Maclaurin's and Taylor's Series. Errors and Approximation.

SUCCESSIVE DIFFERENTIATION

If a function $y = f(x)$ is differentiable we can obtain its derivative and denote it by $f'(x)$. If $f'(x)$ is also differentiable we can get its derivative which is called second order derivative of $f(x)$ and is denoted by $f''(x)$. Continuing this process we get third and higher order derivatives and denote them by $f'''(x)$, $f^{iv}(x)$, ..., $f^{(n)}(x)$, This notation is somewhat cumbersome in practice and a better notation is given by

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \dots, \frac{d^ny}{dx^n}, \dots \text{ or, alternately, the suffix notation } y_1, y_2, y_3, \dots, y_n, \dots$$

where the suffix n is a positive integer giving the order of derivative.

Yet another notation is the operator notation as Dy , D^2y , D^3y , ..., D^ny , ...

where D symbolizes the operation of differentiation, $D \equiv \frac{d}{dx}$, $D^2 \equiv \frac{d^2}{dx^2}$, ..., $D^n \equiv \frac{d^n}{dx^n}$

n^{th} Order Derivatives of some Standard Functions

1. Let $y = e^{ax}$, then $y_1 = ae^{ax}$, $y_2 = a^2e^{ax}$, ..., $y_n = a^ne^{ax}$, that is,

$$\frac{d^n}{dx^n} e^{ax} = a^n e^{ax}$$

2. Let $y = (ax + b)^m$ then, $y_1 = ma(ax + b)^{m-1}$, $y_2 = m(m-1)a^2(ax + b)^{m-2}$, ...

In general, $y_n = m(m-1)(m-2) \dots (m-n+1)a^n(ax + b)^{m-n}$.

In particular, if m is a positive integer greater than n , then we can write,

$$y_n = \frac{d^n}{dx^n} (ax + b)^m = \frac{m!}{(m-n)!} a^n (ax + b)^{m-n}$$

and if m is a positive integer less than n , then $y_n = 0$.

Further, if $m = n$, then $y_n = \frac{d^n}{dx^n} (ax + b)^n = n! a^n$.

and if $m = -1$ $y_n = \frac{d^n}{dx^n} (ax + b)^{-1} = (-1)(-2)(-3) \dots (-n)a^n(ax + b)^{-1-n} = \frac{(-1)^n n! a^n}{(ax + b)^{n+1}}$.

3. Let $y = \log(ax + b)$, then $y_1 = \frac{a}{ax + b} = a(ax + b)^{-1}$

and $y_n = \frac{d^n}{dx^n} \log(ax + b) = \frac{d^{n-1}}{dx^{n-1}} \left(\frac{a}{ax + b} \right) = a \cdot (-1)^{n-1} (n-1)! a^{n-1} (ax + b)^{-1-(n-1)}$

or $y_n = \frac{d^n}{dx^n} \log(ax + b) = (-1)^{n-1} \frac{(n-1)! a^n}{(ax + b)^n}$

4. Let $y = a^x$ then $y_1 = a^x \log a$, $y_2 = a^x (\log a)^2$, ..., $y_n = \frac{d^n}{dx^n} a^x = a^x (\log a)^n$.

5. Let $y = \sin(ax + b)$, then, $y_1 = a \cos(ax + b) = a \sin(ax + b + \pi/2)$
 $y_2 = a^2 \cos(ax + b + \pi/2) = a^2 \sin(ax + b + 2 \cdot \pi/2)$

$$y_n = \frac{d^n}{dx^n} \sin(ax + b) = a^n \sin(ax + b + n\pi/2)$$

Similarly, if $y = \cos(ax + b)$,

then, $y_1 = -a \sin(ax + b) = a \cos(ax + b + \pi/2)$
 $y_2 = -a^2 \sin(ax + b + \pi/2) = a^2 \cos(ax + b + 2 \cdot \pi/2)$

$$y_n = \frac{d^n}{dx^n} \cos(ax + b) = a^n \cos(ax + b + n\pi/2).$$

The above two formulae, in brief, can be written as

$$\frac{d^n}{dx^n} \sin/\cos(ax + b) = a^n \sin/\cos(ax + b + n\pi/2).$$

6. Let $y = e^{ax} \sin(bx + c)$,

then, $y_1 = ae^{ax} \sin(bx + c) + e^{ax} b \cos(bx + c) = e^{ax} [a \sin(bx + c) + b \cos(bx + c)]$

If we put $a = r \cos \alpha$, $b = r \sin \alpha$, then

$$y_1 = e^{ax} [r \sin(bx + c) \cos \alpha + r \cos(bx + c) \sin \alpha] = re^{ax} \sin(bx + c + \alpha)$$

where $r = \sqrt{a^2 + b^2}$ and $\alpha = \tan^{-1}(b/a)$.

Again differentiating and making the above substitution, gives

$$y_2 = r^2 e^{ax} \sin(bx + c + 2\alpha)$$

and continuing the process, in general, we get

$$y_n = r^n e^{ax} \sin(bx + c + n\alpha).$$

Similarly, if $y = e^{ax} \cos(bx + c)$, and proceeding as above, we get

$$y_n = r^n e^{ax} \cos(bx + c + n\alpha).$$

The above two results can be together written as

$$\frac{d^n}{dx^n} e^{ax} \sin/\cos(bx + c) = r^n e^{ax} \sin/\cos(bx + c + n\alpha) \quad \text{where } r = \sqrt{a^2 + b^2} \text{ and } \alpha = \tan^{-1} b/a.$$

EXAMPLE 1.1. (a) Find the n^{th} derivative of $\frac{x^4}{(x-1)(x-2)}$.

[GGSIPU 2011, 2012]

(b) If $y = (2-3x)^{10}$, find y_9 .

[GGSIPU 2013]

SOLUTION: (a) Let $y = \frac{x^4}{(x-1)(x-2)} = x^2 + 3x + 7 + \frac{15x-14}{(x-1)(x-2)}$ {As fraction is improper, }
{convert into proper fraction}

$$= x^2 + 3x + 7 - \frac{1}{x-1} + \frac{16}{x-2}$$

then n^{th} derivative of $y = y_n = 0 - \frac{(1)^n n!}{(x-1)^{n+1}} + \frac{16(-1)^n n!}{(x-2)^{n+1}}, (n \geq 3)$

$$= (-1)^n n! \left[\frac{16}{(x-2)^{n+1}} - \frac{1}{(x-1)^{n+1}} \right] \quad \text{Ans.}$$

(b) $y_9 = \frac{d^9}{dx^9} (2-3x)^{10} = (-3)^9 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 (2-3x)$

$$= 10!(3^9)(-1)^9(2-3x) = 10!(3^9)(3x-2) \quad \text{Ans.}$$

EXAMPLE 1.2. (a) Find the n^{th} derivative of $\sin^2 x \cos^3 x$.

(b) If $y = e^{5x} \sin(2x+3) \sin(x-1) \cos(x+1)$, find y_n .

SOLUTION: (a) Let $y = \sin^2 x \cos^3 x$, we can write

$$y = \left(\frac{1 - \cos 2x}{2} \right) \cdot \left(\frac{\cos 3x + 3 \cos x}{4} \right) = \frac{1}{8} [\cos 3x + 3 \cos x - \cos 2x \cos 3x - 3 \cos 2x \cos x]$$

$$= \frac{1}{8} \left[\cos 3x + 3 \cos x - \frac{1}{2} (\cos 5x + \cos x) - \frac{3}{2} (\cos 3x + \cos x) \right] = \frac{1}{16} [2 \cos x - \cos 3x - \cos 5x]$$

$$\therefore y_n = \frac{1}{16} \left[2 \cos \left(x + \frac{n\pi}{2} \right) - 3^n \cos \left(3x + \frac{n\pi}{2} \right) - 5^n \cos \left(5x + \frac{n\pi}{2} \right) \right] \quad \text{Ans.}$$

(b) We have $y = \frac{1}{2} e^{5x} \sin(2x+3) (\sin 2x - \sin 2) = \frac{1}{2} e^{5x} \{ \sin(2x+3) \sin 2x - \sin(2x+3) \cdot \sin 2 \}$

$$= \frac{1}{4} e^{5x} \{ \cos 3 - \cos(4x+3) \} - \frac{1}{2} \sin 2 \cdot e^{5x} \sin(2x+3)$$

$$= \frac{1}{4} \cos 3 \cdot e^{5x} - \frac{1}{4} e^{5x} \cos(4x+3) - \frac{1}{2} \sin 2 \cdot e^{5x} \sin(2x+3)$$

Therefore, $y_n = \frac{1}{4} \cos 3 \cdot 5^n e^{5x} - \frac{e^{5x}}{4} r_1^n \cos(4x+3+n\alpha) - \frac{1}{2} \sin 2 \cdot r_2^n e^{5x} \sin(2x+3+n\beta)$

where $r_1 = \sqrt{5^2 + 4^2} = \sqrt{41}$, $r_2 = \sqrt{5^2 + 2^2} = \sqrt{29}$, $\alpha = \tan^{-1} \frac{4}{5}$ and $\beta = \tan^{-1} \frac{2}{5}$. Ans.

EXAMPLE 1.3. (a) If $y = \frac{x}{x^2 + a^2}$, find y_n , the n^{th} derivative of y .

(b) Find the n^{th} derivative of $\tan^{-1}(x/a)$.

[GGSIPU 2008]

SOLUTION: (a) Here $y = \frac{x}{(x+ai)(x-ai)} = \frac{1}{2} \left\{ \frac{1}{x-ai} + \frac{1}{x+ai} \right\}$

$$\therefore y_n = \frac{1}{2} \frac{(-1)^n n!}{(x-ai)^{n+1}} + \frac{1}{2} \frac{(-1)^n n!}{(x+ai)^{n+1}}$$

To simplify the above expression, we put $x = r \cos \theta$, $a = r \sin \theta$ so that $\tan \theta = a/x$, and apply De Moivre's theorem and get

$$\begin{aligned} y_n &= \frac{(-1)^n n!}{2} \left[\frac{1}{r^{n+1} (\cos \theta - i \sin \theta)^{n+1}} + \frac{1}{r^{n+1} (\cos \theta + i \sin \theta)^{n+1}} \right] \\ &= \frac{(-1)^n n!}{2 r^{n+1}} [\{\cos(n+1)\theta + i \sin(n+1)\theta\} + \{\cos(n+1)\theta - i \sin(n+1)\theta\}] \\ &= \frac{(-1)^n n!}{r^{n+1}} \cos(n+1)\theta = \frac{(-1)^n n!}{a^{n+1}} \cos(n+1)\theta \sin^{n+1}\theta \quad (\text{using } a = r \sin \theta) \end{aligned}$$

where $\cot \theta = x/a$ or $\theta = \cot^{-1}(x/a) = \tan^{-1}(a/x)$. **Ans.**

(b) Let $y = \tan^{-1}(x/a)$ then $y_1 = \frac{a}{x^2 + a^2}$

or $y_1 = \frac{a}{(x+ai)(x-ai)} = \frac{a}{2ai} \left\{ \frac{1}{x-ai} - \frac{1}{x+ai} \right\}$ where $i = \sqrt{-1}$.

Differentiating the above relation $(n-1)$ times w.r.t. x , gives

$$\begin{aligned} y_n &= \frac{1}{2i} \left\{ \frac{d^{n-1}}{dx^{n-1}} \frac{1}{x-ai} - \frac{d^{n-1}}{dx^{n-1}} \frac{1}{x+ai} \right\} = \frac{1}{2i} \left\{ \frac{(-1)^{n-1} (n-1)!}{(x-ai)^n} - \frac{(-1)^{n-1} (n-1)!}{(x+ai)^n} \right\} \\ &= \frac{(-1)^{n-1} (n-1)!}{2i} \left\{ \frac{1}{(x-ai)^n} - \frac{1}{(x+ai)^n} \right\}. \end{aligned}$$

The above expression can be simplified by using De Moivre's theorem, for which we put $x = r \cos \theta$, $a = r \sin \theta$, then

$$\begin{aligned} y_n &= \frac{(-1)^{n-1} (n-1)!}{2i} \left[\frac{1}{r^n (\cos \theta - i \sin \theta)^n} - \frac{1}{r^n (\cos \theta + i \sin \theta)^n} \right] \\ &= \frac{(-1)^{n-1} (n-1)!}{2i r^n} [(\cos \theta - i \sin \theta)^{-n} - (\cos \theta + i \sin \theta)^{-n}] \\ &= \frac{(-1)^{n-1} (n-1)!}{2i r^n} [\cos n\theta + i \sin n\theta - \cos n\theta + i \sin n\theta] = \frac{(-1)^{n-1} (n-1)!}{r^n} \sin n\theta \\ &= \frac{(-1)^{n-1} (n-1)!}{(a/\sin \theta)^n} \sin n\theta = \frac{(-1)^{n-1} (n-1)!}{a^n} \sin n\theta \sin^n \theta \quad (\text{using } a = r \sin \theta) \end{aligned}$$

where $\tan \theta = \frac{a}{x}$ or $\theta = \tan^{-1}(a/x) = \cot^{-1}(x/a)$. **Ans.**

EXAMPLE 1.4. If $p^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta$, show that $p + \frac{d^2 p}{d\theta^2} = \frac{a^2 b^2}{p^3}$.

SOLUTION: Differentiating $p^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta$ we get $2p \frac{dp}{d\theta} = -2a^2 \cos \theta \sin \theta + 2b^2 \sin \theta \cos \theta \dots (1)$

or $p \frac{dp}{d\theta} = (b^2 - a^2) \sin \theta \cos \theta \dots (2)$

Differentiating w.r.t. θ again, we get $p \frac{d^2 p}{d\theta^2} + \left(\frac{dp}{d\theta}\right)^2 = (b^2 - a^2)(\cos^2 \theta - \sin^2 \theta)$

Multiplying throughout by p^2 and using (2), gives

$$\begin{aligned} p^3 \frac{d^2 p}{d\theta^2} + (b^2 - a^2)^2 \sin^2 \theta \cos^2 \theta &= (b^2 - a^2) p^2 (\cos^2 \theta - \sin^2 \theta) \\ &= p^2 \{ (b^2 \cos^2 \theta + a^2 \sin^2 \theta) - (a^2 \cos^2 \theta + b^2 \sin^2 \theta) \} \\ &= p^2 (b^2 \cos^2 \theta + a^2 \sin^2 \theta) - p^4 \end{aligned}$$

$$\begin{aligned} \text{or } p^3 \frac{d^2 p}{d\theta^2} + p^4 &= (a^2 \cos^2 \theta + b^2 \sin^2 \theta) (b^2 \cos^2 \theta + a^2 \sin^2 \theta) - (b^2 - a^2)^2 \sin^2 \theta \cos^2 \theta \\ &= a^2 b^2 (\cos^4 \theta + \sin^4 \theta) + (a^4 + b^4) \sin^2 \theta \cos^2 \theta - (a^4 + b^4 - 2a^2 b^2) \sin^2 \theta \cos^2 \theta \\ &= a^2 b^2 (\cos^4 \theta + \sin^4 \theta + 2 \sin^2 \theta \cos^2 \theta) = a^2 b^2 (\cos^2 \theta + \sin^2 \theta)^2 = a^2 b^2 \end{aligned}$$

$$\text{or } \frac{d^2 p}{d\theta^2} + p = \frac{a^2 b^2}{p^3}. \quad \text{Hence Proved.}$$

EXAMPLE 1.5. (a) If $y = \log(x + \sqrt{1+x^2})$ prove that $(1+x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = 0$. [GGSIPU 2009]

(b) If $y = x \log \left(\frac{x-1}{x+1} \right)$, show that $y_n = (-1)^{n-2} (n-2)! \left[\frac{x-n}{(x-1)^n} - \frac{x+n}{(x+1)^n} \right]$.

SOLUTION: (a) $y = \log(x + \sqrt{1+x^2})$ hence $\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}}$

or $(\sqrt{1+x^2}) \frac{dy}{dx} = 1$. Differentiating w.r.t. x , on both sides, we get

$$\sqrt{1+x^2} \frac{d^2 y}{dx^2} + \frac{2x}{2\sqrt{1+x^2}} \frac{dy}{dx} = 0 \quad \text{or} \quad (1+x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = 0. \quad \text{Hence proved.}$$

(b) Here $y = x \log \left(\frac{x-1}{x+1} \right) = x \log(x-1) - x \log(x+1)$

$$\therefore y_1 = \frac{x}{x-1} + \log(x-1) - \frac{x}{x+1} - \log(x+1) = \frac{1}{x-1} + \frac{1}{x+1} + \log(x-1) - \log(x+1)$$

Differentiating both sides $(n-1)$ times with respect to x , we get

$$\begin{aligned} y_n &= \frac{(-1)^{n-1} (n-1)!}{(x-1)^n} + \frac{(-1)^{n-1} (n-1)!}{(x+1)^n} + \frac{(-1)^{n-2} (n-2)!}{(x-1)^{n-1}} - \frac{(-1)^{n-2} (n-2)!}{(x+1)^{n-1}} \\ &= (-1)^{n-2} (n-2)! \left\{ \frac{-(n-1) + x - 1}{(x-1)^n} \right\} + (-1)^{n-2} (n-2)! \left\{ \frac{-(n-1) - (x+1)}{(x+1)^n} \right\} \\ &= (-1)^{n-2} (n-2)! \left[\frac{x-n}{(x-1)^n} - \frac{x+n}{(x+1)^n} \right] \end{aligned}$$

Hence the result.

EXAMPLE 1.6. (a) Prove that the value of n^{th} derivative of $\frac{x^3}{x^2-1}$ at $x=0$, is 0 if n is even and $-n!$ if n is odd and greater than 1.

(b) Find the n^{th} derivative of $\frac{1}{1+x+x^2}$.

[GGSIPU 2007]

SOLUTION: (a) Let $y = \frac{x^3}{x^2-1} = x + \frac{x}{x^2-1} = x + \frac{1}{2} \left(\frac{1}{x-1} \right) + \frac{1}{2} \left(\frac{1}{x+1} \right)$ obviously $y'(0) = 0$.

Then $y_n = 0 + \frac{1}{2} \frac{d^n}{dx^n} \frac{1}{x-1} + \frac{1}{2} \frac{d^n}{dx^n} \frac{1}{x+1}$ when $n > 1$. $= \frac{1}{2} \frac{(-1)^n n!}{(x-1)^{n+1}} + \frac{1}{2} \frac{(-1)^n n!}{(x+1)^{n+1}}$.

$$\begin{aligned} \therefore y_n(0) &= \frac{(-1)^n n!}{2} \left[\frac{1}{(-1)^{n+1}} + 1 \right] = \frac{(-1)^n n!}{2(-1)^{n+1}} [1 + (-1)^{n+1}] \\ &= \frac{-n!}{2} [1 + (-1)^{n+1}] = \begin{cases} 0 & \text{if } n \text{ is even} \\ -n! & \text{if } n \text{ is odd} \end{cases} \end{aligned}$$

Hence Proved.

(b) $y = \frac{1}{1+x+x^2} = \frac{1}{(x-\omega)(x-\omega^2)}$ where $\omega = \frac{-1+i\sqrt{3}}{2}$, $\omega^2 = \frac{-1-i\sqrt{3}}{2}$.

$$= \left(\frac{1}{x-\omega} - \frac{1}{x-\omega^2} \right) \frac{1}{\omega-\omega^2} = \frac{-i}{\sqrt{3}} \left(\frac{1}{x-\omega} - \frac{1}{x-\omega^2} \right)$$

$$\begin{aligned} \therefore y_n &= \frac{d^n}{dx^n} \frac{1}{1+x+x^2} = \frac{-i}{\sqrt{3}} \left[\frac{d^n}{dx^n} \frac{1}{x-\omega} - \frac{d^n}{dx^n} \frac{1}{x-\omega^2} \right] \\ &= \frac{-i}{\sqrt{3}} \left[\frac{(-1)^n n!}{(x-\omega)^{n+1}} - \frac{(-1)^n n!}{(x-\omega^2)^{n+1}} \right] = i(-1)^{n+1} \frac{n!}{\sqrt{3}} \left[\frac{1}{(x-\omega)^{n+1}} - \frac{1}{(x-\omega^2)^{n+1}} \right] \\ &= \frac{(-1)^{n+1}}{\sqrt{3}} n! i \left[\frac{1}{\left(x + \frac{1}{2} - \frac{i\sqrt{3}}{2} \right)^{n+1}} - \frac{1}{\left(x + \frac{1}{2} + \frac{i\sqrt{3}}{2} \right)^{n+1}} \right] \\ &= (-1)^{n+1} \frac{2^{n+1}}{\sqrt{3}} n! i \left[\frac{1}{(2x+1-i\sqrt{3})^{n+1}} - \frac{1}{(2x+1+i\sqrt{3})^{n+1}} \right] \end{aligned}$$

Putting $2x+1 = r \cos \theta$, $\sqrt{3} = r \sin \theta$, we get

$$\begin{aligned} y_n &= \frac{(-1)^{n+1} 2^{n+1} n! i}{\sqrt{3} \cdot r^{n+1}} [(\cos \theta - i \sin \theta)^{-(n+1)} - (\cos \theta + i \sin \theta)^{-(n+1)}] \\ &= \frac{(-1)^{n+1} 2^{n+1} n! i}{\sqrt{3} \left(\frac{\sqrt{3}}{\sin \theta} \right)^{n+1}} [\cos(n+1)\theta + i \sin(n+1)\theta - \cos(n+1)\theta + i \sin(n+1)\theta] \\ &= \frac{(-1)^{n+1} 2^{n+1} n! \cdot i \sin^{n+1} \theta}{(\sqrt{3})^{n+2}} \cdot 2i \sin(n+1)\theta \\ &= (-1)^{n+2} \frac{2^{n+2} n!}{(\sqrt{3})^{n+2}} \sin^{n+1} \theta \sin(n+1)\theta = \frac{(-1)^n n!}{a^{n+2}} \sin^{n+1} \theta \sin(n+1)\theta \end{aligned}$$

where $\tan \theta = \frac{\sqrt{3}}{2x+1}$ and $a = \frac{\sqrt{3}}{2}$. **Ans.**

LEIBNITZ'S THEOREM

This theorem provides the n th derivative of the product of two functions. It states as follows:

If $y = u \cdot v$ where u and v are functions of x and their n th derivatives are separately known, then

$$y_n = (uv)_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_r u_{n-r} v_r + \dots + u \cdot v_n$$

Proof : The theorem will be proved by the principle of Mathematical induction.

$$y_1 = \frac{du}{dx} v + u \frac{dv}{dx} = u_1 v + u v_1 \text{ which shows that the theorem is true for } n = 1.$$

Now, let us suppose that the statement of the theorem is true for $n = K$, i.e. suppose

$$(uv)_K = u_K v + {}^KC_1 u_{K-1} v_1 + {}^KC_2 u_{K-2} v_2 + \dots + u v_K \text{ is true.}$$

Then differentiating both sides w.r.t. x , gives

$$\begin{aligned} (uv)_{K+1} &= (u_{K+1} v + u_K v_1) + {}^KC_1 (u_K v_1 + u_{K-1} v_2) + {}^KC_2 (u_{K-1} v_2 + u_{K-2} v_3) + \dots \\ &\quad + (u_1 v_K + u v_{K+1}) \\ &= u_{K+1} v + (1 + {}^KC_1) u_K v_1 + ({}^KC_1 + {}^KC_2) u_{K-1} v_2 + \dots + ({}^KC_{K-1} + {}^KC_K) u_1 v_K \\ &\quad + \dots + u v_{K+1} \\ &= u_{K+1} v + {}^{K+1}C_1 u_K v_1 + {}^{K+1}C_2 u_{K-1} v_2 + \dots + {}^{K+1}C_K u_1 v_K + u v_{K+1} \\ &\quad \left(\text{on using the result } {}^KC_r + {}^KC_{r+1} = {}^{K+1}C_{r+1} \text{ and } 1 = {}^KC_0 \right) \end{aligned}$$

This shows that the statement of the theorem is true for $n = K + 1$ under the supposition that it is true for $n = K$. Therefore, by the principle of Mathematical induction the theorem is true for every positive integral value of n .

EXAMPLE 1.7. (a) If $y = x^2 e^x$ show that $y_n = \frac{n(n-1)}{2} y_2 - n(n-2) y_1 + \frac{(n-1)(n-2)}{2} y$.

[GGSIPU 2004]

(b) Find the n th derivative of $x^2 e^{3x} \sin 4x$.

SOLUTION: (a) $y = x^2 e^x$ so $y_1 = 2xe^x + x^2 e^x = 2xe^x + y$

Differentiating again, we get $y_2 = 2e^x + 2e^x + y_1$ or $2e^x = y_2 - 2y_1 + y$.

Now applying Leibnitz theorem on $x^2 e^x$, we get

$$\begin{aligned} y_n &= x^2 e^x + n \cdot 2xe^x + \frac{n(n-1)}{2} \cdot 2e^x = y + n(y_1 - y) + \frac{n(n-1)}{2} (y_2 - 2y_1 + y) \\ &= \frac{n(n-1)}{2} y_2 + y_1 [n - n(n-1)] + y \left[1 - n + \frac{n(n-1)}{2} \right] \\ &= \frac{n(n-1)}{2} y_2 - n(n-2) y_1 + \frac{1}{2} (n-1)(n-2) y. \end{aligned}$$

Hence the result.

(b) Let $y = u v$ where $u = e^{3x} \sin 4x$ and $v = x^2$.

Hence, $u_n = e^{3x} r^n \sin(4x + n\alpha)$ where $r = \sqrt{3^2 + 4^2} = 5$ and $\tan \alpha = 4/3$,

Here $v_1 = 2x$, $v_2 = 2$, $v_3 = v_4 = \dots = v_n = 0$.

Therefore, by Leibnitz's theorem

$$\begin{aligned} y_n &= u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 \text{ as } v_3 = v_4 = \dots = 0; \quad v = x^2 \\ &= 5^n e^{3x} \sin(4x + n\alpha) x^2 + {}^nC_1 5^{n-1} e^{3x} \sin(4x + (n-1)\alpha) (2x) + {}^nC_2 5^{n-2} e^{3x} \sin(4x + (n-2)\alpha) \cdot (2) \end{aligned}$$

$$= 5^n e^{3x} \left[x^2 \sin(4x + n\alpha) + \frac{2nx}{5} \sin(4x + (n-1)\alpha) + \frac{n(n-1) \cdot (2)}{2 \cdot 25} \sin(4x + (n-2)\alpha) \right]$$

$$\text{or } y_n = 5^n e^{3x} \left[x^2 \sin(4x + n\alpha) + \frac{2nx}{5} \sin(4x + n\alpha - \alpha) + \frac{n(n-1)}{25} \sin(4x + n\alpha - 2\alpha) \right]$$

where $\alpha = \tan^{-1}(4/3)$.

Ans.

EXMAPLE 1.8. (a) If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, prove that

$$(1-x^2)y_{n+1} - (2n+1)xy_n - n^2y_{n-1} = 0 \text{ and hence find } (y_n)_0. \quad [\text{GGSIPU 2010}]$$

(b) If $\log y = \tan^{-1} x$, show that $(1+x^2)y_{n+2} + \{2(n+1)x-1\}y_{n+1} + n(n+1)y_n = 0$
and hence find y_3, y_4 and y_5 at $x=0$. [GGSIPU 2009; 2014]

SOLUTION: (a) The given relation can be written as $y\sqrt{1-x^2} = \sin^{-1} x \quad \dots(1)$

On differentiating w.r.t. x on both sides of (1), gives

$$\sqrt{1-x^2}y_1 - \frac{2x}{2\sqrt{1-x^2}}y = \frac{1}{\sqrt{1-x^2}} \quad \text{or} \quad (1-x^2)y_1 - xy = 1 \quad \dots(2)$$

Differentiating (2) n -times using Leibnitz's theorem, we get

$$(1-x^2)y_{n+1} + {}^nC_1(-2x)y_n + {}^nC_2(-2)y_{n-1} - xy_n - {}^nC_1 \cdot 1 \cdot y_{n-1} = 0$$

$$\text{or } (1-x^2)y_{n+1} - (2n+1)xy_n - n^2y_{n-1} = 0,$$

hence the result.

At $x=0$ above relation becomes $(y_{n+1})_0 = n^2(y_{n-1})_0$

$$\text{or } (y_n)_0 = (n-1)^2(y_{n-2})_0 = (n-1)^2(n-3)^2(y_{n-4})_0 = (n-1)^2(n-3)^2(n-5)^2(y_{n-6})_0 = \dots$$

$$\Rightarrow (y_n)_0 = \begin{cases} (n-1)^2(n-3)^2(n-5)^2 \dots 3^2 \cdot 1^2 (y)_0 & \text{when } n \text{ is even} \\ (n-1)^2(n-3)^2(n-5)^2 \dots 4^2 \cdot 2^2 (y_1)_0 & \text{when } n \text{ is odd} \end{cases}$$

Since $y(0) = 0$ we have $(y_n)_0 = 0$ when n is even,

and, since from (2) $(y_1)_0 = 1$, we have $(y_n)_0 = (n-1)^2(n-3)^2(n-5)^2 \dots 2^2 \cdot 1$ when n is odd. **Ans.**

$$(b) \text{ We have } \log y = \tan^{-1} x \quad \text{or} \quad y = e^{\tan^{-1} x} \quad \dots(1)$$

$$\text{hence, } y_1 = e^{\tan^{-1} x} \cdot \frac{1}{1+x^2} \quad \text{or} \quad (1+x^2)y_1 = y \quad \dots(2)$$

Differentiating (2) $(n+1)$ times using Leibnitz's theorem, gives

$$(1+x^2)y_{n+2} + {}^{n+1}C_1 \cdot 2xy_{n+1} + {}^{n+1}C_2 \cdot 2y_n = y_{n+1}$$

$$\text{or } (1+x^2)y_{n+2} + \{2(n+1)x-1\}y_{n+1} + n(n+1)y_n = 0. \quad \dots(3)$$

$$\text{At } x=0 \text{ relation (3) becomes } (y_{n+2})_0 - (y_{n+1})_0 + n(n+1)(y_n)_0 = 0. \quad \dots(4)$$

$$\text{Taking } n=0 \text{ in (4), we get } (y_2)_0 - (y_1)_0 + 0(y)_0 = 0 \quad \text{or} \quad (y_2)_0 = (y_1)_0$$

But from (2), $(y_1)_0 = y(0) = 1$ hence $(y_2)_0 = 1$

Taking $n = 1$ in (4), we get $(y_3)_0 - (y_2)_0 + 2(y_1)_0 = 0 \Rightarrow (y_3)_0 = -1$

Taking $n = 2$ in (4), we get $(y_4)_0 - (y_3)_0 + 2.3(y_2)_0 = 0 \Rightarrow (y_4)_0 = -7$

and lastly, taking $n = 3$ in (4), we get $(y_5)_0 - (y_4)_0 + 3.4(y_3)_0 = 0 \Rightarrow (y_5)_0 = 5$. **Ans.**

EXMAPLE 1.9. (a) If $x + y = 1$ and $D \equiv \frac{d}{dx}$, prove that

$$D^n(x^n y^n) = n! \left[y^n - \binom{n}{1}^2 y^{n-1} x + \binom{n}{2}^2 y^{n-2} x^2 - \dots + (-1)^n x^n \right]$$

(b) If $V_n = \frac{d^n}{dx^n}(x^n \log x)$, establish $V_n = n V_{n-1} + (n-1)!$

and hence show that $V_n = n! \left(\log x + 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right)$.

[AKTU 2017; GGSIPU 2017]

SOLUTION: (a) Here $x + y = 1$ therefore, $\frac{dy}{dx} = Dy = -1$.

Applying Leibnitz's theorem, to $D^n(x^n y^n)$, we get

$$D^n(x^n y^n) = y^n D^n(x^n) + {}^n C_1 D(y^n) D^{n-1}(x^n) + {}^n C_2 D^2(y^n) D^{n-2}(x^n) + \dots + D^n(y^n) \cdot x^n.$$

$$= y^n n! + {}^n C_1 \cdot n y^{n-1} (-1) \cdot n! x + {}^n C_2 n(n-1) (-1)^2 y^{n-2} n! \frac{x^2}{2!} + \dots + n! (-1)^n x^n$$

$$= n! \left[y^n - {}^n C_1 \cdot n y^{n-1} x + {}^n C_2 \cdot \frac{n(n-1)}{2!} y^{n-2} x^2 - \dots + (-1)^n x^n \right]$$

$$\text{or } D^n(x^n y^n) = n! \left[y^n - \binom{n}{1}^2 y^{n-1} x + \binom{n}{2}^2 y^{n-2} x^2 - \binom{n}{3}^2 y^{n-3} x^3 + \dots + (-1)^n x^n \right].$$

hence the result.

$$\begin{aligned} \text{(b)} \quad V_n &= \frac{d^n}{dx^n}(x^n \log x) = \frac{d^{n-1}}{dx^{n-1}} \left\{ \frac{d}{dx}(x^n \log x) \right\} = \frac{d^{n-1}}{dx^{n-1}} \left\{ x^n \frac{1}{x} + n x^{n-1} \log x \right\} \\ &= \frac{d^{n-1}}{dx^{n-1}} x^{n-1} + n \frac{d^{n-1}}{dx^{n-1}} (x^{n-1} \log x) = (n-1)! + n V_{n-1} \end{aligned} \quad \dots(1)$$

Dividing throughout by $n!$, we get

$$\frac{V_n}{n!} = \frac{(n-1)!}{n!} + \frac{n}{n!} V_{n-1} = \frac{1}{n} + \frac{V_{n-1}}{(n-1)!}. \quad \dots(2)$$

Replacing n by $n-1$ in the above relation, we get $\frac{V_{n-1}}{(n-1)!} = \frac{1}{n-1} + \frac{V_{n-2}}{(n-2)!}$

$\therefore$ (2) becomes $\frac{V_n}{n!} = \frac{1}{n} + \frac{1}{n-1} + \frac{V_{n-2}}{(n-2)!}$ (again using (2) replacing n by $n-2$, we have)

$$= \frac{1}{n} + \frac{1}{n-1} + \frac{1}{n-2} + \frac{V_{n-3}}{(n-3)!}$$

$$= \dots$$

$$= \dots$$

$$= \frac{1}{n} + \frac{1}{n-1} + \frac{1}{n-2} + \frac{1}{n-3} + \dots + \frac{1}{2} + \frac{1}{1} + \frac{V_0}{0!} \quad \text{But } V_0 = x^0 \log x = \log x$$

$$\text{Therefore, } V_n = n! \left[\log x + 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right].$$

Hence Proved.

EXAMPLE 1.10. (a) If $y = e^{x^2/2} \cos x$, show that $y_{n+2}(0) - 2n y_n(0) + n(n-1) y_{n-2}(0) = 0$.

(b) If $y = \sqrt{\frac{1+x}{1-x}}$ prove that $y = (1-x^2) y_1$ and hence show that

$$(1-x^2) y_n - \{2(n-1)x + 1\} y_{n-1} - (n-1)(n-2) y_{n-2} = 0$$

SOLUTION: (a) We have $y = e^{x^2/2} \cos x$ which on differentiation, gives

$$y_1 = e^{x^2/2} x \cos x - e^{x^2/2} \sin x = xy - e^{x^2/2} \sin x$$

$$\text{and } y_2 = xy_1 + y - e^{x^2/2} x \sin x - e^{x^2/2} \cos x \quad \text{or} \quad y_2 = 2xy_1 - x^2 y.$$

Now, differentiating the above relation n times and using Leibnitz's theorem, gives

$$y_{n+2} = 2xy_{n+1} + {}^nC_1 \cdot 2 \cdot y_n - (x^2 y_n + {}^nC_1 2xy_{n-1} + {}^nC_2 2y_{n-2})$$

$$\text{or } y_{n+2} - 2xy_{n+1} + (x^2 - 2n) y_n + 2nxy_{n-1} + n(n-1) y_{n-2} = 0$$

Putting $x = 0$ throughout, we get $y_{n+2}(0) - 2n y_n(0) + n(n-1) y_{n-2}(0) = 0$, hence the result. ...(1)

(b) The given relation can be written as $(1-x)y^2 = 1+x$

This, on differentiation w.r.t. x , gives $(1-x) 2yy_1 - y^2 = 1$

$$\text{or } (1-x) 2yy_1 = 1 + y^2 = 1 + \frac{1+x}{1-x} = \frac{2}{1-x}$$

$$\text{Multiplying throughout by } y, \text{ gives } (1-x) 2y^2 y_1 = \frac{2y}{1-x}.$$

$$\text{Using (1) here, we get } (1-x)^2 \left(\frac{1+x}{1-x} \right) y_1 = y \quad \text{or} \quad (1-x^2) y_1 = y$$

Hence Proved.

Now, differentiating this relation n times, with respect to x and using Leibnitz's theorem, we get

$$(1-x^2) y_{n+1} + {}^nC_1 (-2x) y_n + {}^nC_2 (-2) y_{n-1} = y_n$$

$$\text{or } (1-x^2) y_{n+1} - (2nx + 1) y_n - n(n-1) y_{n-1} = 0$$

Replacing n by $n-1$ here, gives

$$(1-x^2) y_n - \{2(n-1)x + 1\} y_{n-1} - (n-1)(n-2) y_{n-2} = 0 \quad \text{hence the desired result.}$$

EXAMPLE 1.11. If $y^{1/m} + y^{-1/m} = 2x$ show that $(x^2 - 1)y_{n+2} + (2n+1)x y_{n+1} + (n^2 - m^2)y_n = 0$.

[GGSIPU 2004; 2011][AKTU 2015]

SOLUTION: The given relation can be written as

$$y^{2/m} - 2xy^{1/m} + 1 = 0 \quad \text{whose roots are } y^{1/m} = \frac{2x \pm \sqrt{4x^2 - 4}}{2}$$

$$\text{or } y = (x \pm \sqrt{x^2 - 1})^m \quad \text{or } \log y = m \log((x \pm \sqrt{x^2 - 1}))$$

$$\text{Differentiating w.r.t. } x, \text{ we get } \frac{1}{y} y_1 = \frac{m}{x \pm \sqrt{x^2 - 1}} \left[1 \pm \frac{x}{\sqrt{x^2 - 1}} \right] = \pm \frac{m}{\sqrt{x^2 - 1}}$$

or $(x^2 - 1) y_1^2 = m^2 y^2$. This on further differentiation w.r.t. x , gives

$$(x^2 - 1) 2y_1 y_2 + 2x y_1^2 = 2m^2 y y_1 \quad \text{or} \quad (x^2 - 1) y_2 + x y_1 = m^2 y, \text{ if } y_1 \neq 0.$$

Differentiating the above relation n times, using Leibnitz theorem, we get

$$(x^2 - 1) y_{n+2} + n \cdot 2x y_{n+1} + \frac{n(n-1)}{2} \cdot 2 \cdot y_n + x y_{n+1} + n \cdot y_n = m^2 y_n$$

$$\text{or } (x^2 - 1) y_{n+2} + (2n+1)x y_{n+1} + (n^2 - m^2) y_n = 0$$

Hence the result.

- EXAMPLE 1.12.** (a) If $y = (\sin^{-1}x)^2$, prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$ and hence find $y_n(0)$. [GGSIPU 2011; 2014; 2015, 2018][AKTU 2018]
- (b) If $y = e^{m \sin^{-1}x}$, show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (m^2+n^2)y_n = 0$ and hence find $y_n(0)$. [GGSIPU 2011, 2012][AKTU 2016]

SOLUTION: (a) We are given $y = (\sin^{-1}x)^2$ hence $y_1 = \frac{2 \sin^{-1}x}{\sqrt{1-x^2}}$ —(1)

or $(1-x^2)y_1^2 = 4y$ which on differentiation again gives

$$(1-x^2)2y_1y_2 - 2xy_1^2 = 4y_1 \quad \text{or } (1-x^2)y_2 - xy_1 = 2 \quad \text{---(2)}$$

Now differentiating (2) n times w.r.t. x using Leibnitz's theorem, gives

$$(1-x^2)y_{n+2} + n(-2x)y_{n+1} + \frac{n(n-1)}{2}(-2)y_n - xy_{n+1} - n \cdot y_n = 0$$

$$\text{or } (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0 \quad \text{---(3)}$$

Next, at $x = 0$ relation (3) becomes $y_{n+2}(0) = n^2y_n(0)$ —(4)

$$\begin{aligned} \text{or } y_n(0) &= (n-2)^2y_{n-2}(0) = (n-2)^2(n-4)^2y_{n-4}(0) = (n-2)^2(n-4)^2(n-6)^2y_{n-6}(0) = \dots \\ &= \begin{cases} (n-2)^2(n-4)^2(n-6)^2 \dots 2^2y_2(0) & \text{when } n \text{ is even} \\ (n-2)^2(n-4)^2(n-6)^2 \dots 1^2y_1(0) & \text{when } n \text{ is odd} \end{cases} \end{aligned}$$

But from (1), $y_1(0) = 0$ and from (2), $y_2(0) = 2$, therefore

$$y_n(0) = \begin{cases} 0 & \text{when } n \text{ is odd} \\ (n-2)^2(n-4)^2(n-6)^2 \dots 2^2 \cdot 2 & \text{when } n \text{ is even.} \end{cases} \quad \text{Ans.}$$

(b) Given $y = e^{m \sin^{-1}x}$ or $\log y = m \sin^{-1}x$. Differentiating this relation w.r.t. x , we get

$$\frac{1}{y}y_1 = \frac{m}{\sqrt{1-x^2}} \quad \text{or } y_1^2(1-x^2) = m^2y^2 \quad \text{---(1)}$$

On further differentiation, we get $(1-x^2)2y_1y_2 - 2xy_1^2 = 2m^2yy_1$

$$\text{or } (1-x^2)y_2 - xy_1 - m^2y = 0 \quad \text{if } y_1 \neq 0. \quad \text{---(2)}$$

Differentiating the above relation n times, using Leibnitz's theorem, gives

$$(1-x^2)y_{n+2} + n(-2x)y_{n+1} + \frac{n(n-1)}{2}(-2)y_n - xy_{n+1} - n \cdot 1y_n - m^2y_n = 0$$

$$\text{or } (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (m^2+n^2)y_n = 0. \quad \text{---(3)}$$

At $x = 0$ the above relation becomes $y_{n+2}(0) = (m^2+n^2)y_n(0)$

$$\begin{aligned} \Rightarrow y_n(0) &= [m^2+(n-2)^2]y_{n-2}(0) = [m^2+(n-2)^2][m^2+(n-4)^2]y_{n-4}(0) = \dots \\ &= \begin{cases} [m^2+(n-2)^2][m^2+(n-4)^2][m^2+(n-6)^2] \dots [m^2+1]y_1(0) & \text{when } n \text{ is odd} \\ [m^2+(n-2)^2][m^2+(n-4)^2][m^2+(n-6)^2] \dots [m^2+2^2]y_2(0) & \text{when } n \text{ is even} \end{cases} \end{aligned}$$

But from (1) $y_1(0) = my(0) = m$ and from (2), $y_2(0) = m^2 y(0) = m^2$

$$\therefore y_n(0) = \begin{cases} [m^2 + (n-2)^2][m^2 + (n-4)^2][m^2 + (n-6)^2] \dots [m^2 + 1]m & \text{when } n \text{ is odd} \\ [m^2 + (n-2)^2][m^2 + (n-4)^2][m^2 + (n-6)^2] \dots [m^2 + 2^2]m^2 & \text{when } n \text{ is even} \end{cases}$$

Ans.

EXAMPLE 1.13. If $y = \cos(m \sin^{-1} x)$ (or $= \sin(m \sin^{-1} x)$), show that
 $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} + (m^2 - n^2)y_n = 0$ and hence find $y_n(0)$.
 [GGSIPU 2009; 2013][AKTU 2016, 2019]

SOLUTION: $y = \cos(m \sin^{-1} x)$, $\therefore y_1 = \sin(m \sin^{-1} x) \cdot \frac{-m}{\sqrt{1-x^2}}$... (i)

or $(1-x^2)y_1^2 = m^2(1-y^2)$. Again differentiation gives
 $(1-x^2)2y_1 y_2 - 2xy_1^2 = -2m^2 y y_1$ or $(1-x^2)y_2 - xy_1 + m^2 y = 0$ (provided $y_1 \neq 0$) ... (ii)

Differentiating (2) n times, using Leibnitz theorem, gives

or $(1-x^2)y_{n+2} + n(-2x)y_{n+1} + \frac{n(n-1)}{2}(-2)y_n - xy_{n+1} - n \cdot 1 \cdot y_n + m^2 y_n = 0$
 $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$ which is the required result.

Now to find $y_n(0)$ we put $x = 0$ here and get

$y_{n+2}(0) = (n^2 - m^2)y_n(0)$ or $y_n(0) = [(n-2)^2 - m^2]y_{n-2}(0)$... (4)
 $\therefore y_n(0) = [(n-2)^2 - m^2][(n-4)^2 - m^2]y_{n-4}(0)$

$$= \begin{cases} [(n-2)^2 - m^2][(n-4)^2 - m^2] \dots (1^2 - m^2) y_1(0) & \text{if } n \text{ is odd} \\ [(n-2)^2 - m^2][(n-4)^2 - m^2] \dots (2^2 - m^2) y_2(0) & \text{if } n \text{ is even} \end{cases}$$

From (1) we have $y_1(0) = 0$ and from (2) $y_2(0) = -m^2 y(0) = -m^2$.

Therefore $y_n(0) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ [(n-2)^2 - m^2][(n-4)^2 - m^2] \dots [2^2 - m^2](-m^2) & \text{if } n \text{ is even} \end{cases}$

However, if $y = \sin(m \sin^{-1} x)$ then

$y_n(0) = \begin{cases} 0 & \text{if } n \text{ is even} \\ [(n-2)^2 - m^2][(n-4)^2 - m^2] \dots [1^2 - m^2](m) & \text{if } n \text{ is odd} \end{cases}$ Ans.